RAJ’S DSA INTERVIEW ULTIMATE HANDBOOK 2

Complete Software Engineering Preparation Guide

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MOST ASKED TOPICS INCLUDED

- **Recursion & Backtracking**
- **Stack Queue Deque**
- **Sliding Window**
- **Two Pointer**
- **Heap**

Strictly designed for high-impact technical interview preparation.

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MOST ASKED TOPICS

Note - Learn Below Patterns to Crack Any Interview

MOST ASKED

RECURSION & BACKTRACKING

Keywords:

Generate All, Choose, Explore, Undo, Backtrack, Subsets, Combinations, Permutations, Partition, All Possible, DFS, Every Possibility, Subsequences, Path

Identifier:

When the problem asks you to generate all possible answers, explore different choices, or repeatedly make a decision and move deeper.

Hint:

Jab saare possible answers generate karne ho... Recursion socho.

Pattern:

Take → Not Take
Choose → Explore → Undo

What to Look For:

All possible combinations, All subsets, All permutations, Generate all valid strings, Choose / Skip, Partition, Multiple choices, Constraint checking

NOTE - Learn to Calculate Time Complexity of all below Questions

Questions

Pow(X,N)

https://leetcode.com/problems/powx-n/description/

Sort/Reverse Stack using Recursion

https://www.geeksforgeeks.org/dsa/sort-a-stack-using-recursion/

Subsets

https://leetcode.com/problems/subsets/

Combinations

https://leetcode.com/problems/combinations/

Permutations

https://leetcode.com/problems/permutations/

Letter Combinations of a Phone Number

https://leetcode.com/problems/letter-combinations-of-a-phone-number/

Generate Parentheses

https://leetcode.com/problems/generate-parentheses/

Combination Sum

https://leetcode.com/problems/combination-sum/

Combination Sum II

https://leetcode.com/problems/combination-sum-ii/

Subsets II

https://leetcode.com/problems/subsets-ii/

Permutations II

https://leetcode.com/problems/permutations-ii/

BackTracking

Palindrome Partitioning

https://leetcode.com/problems/palindrome-partitioning/

Word Search

https://leetcode.com/problems/word-search/

N-Queens

https://leetcode.com/problems/n-queens/

Sudoku Solver

https://leetcode.com/problems/sudoku-solver/

Partition to K Equal Sum Subsets

https://leetcode.com/problems/partition-to-k-equal-sum-subsets/

STACK / QUEUE / DEQUE

Keywords:
LIFO, FIFO, Stack, Queue, Deque, Monotonic Stack, Monotonic Queue, Next Greater, Next Smaller, Previous Greater, Previous Smaller, Min Stack, LRU, LFU

Identifier:

Previous / Next Greater or Smaller → Stack
First In First Out → Queue
Sliding Window Maximum / Minimum → Deque
LIFO → Stack
FIFO → Queue
Both Ends → Deque
Increasing / Decreasing → Monotonic Stack

What to Look For:

Next Greater, Next Smaller, Previous Greater, Previous Smaller, Parentheses, Expression Evaluation, Largest Rectangle, Sliding Window Maximum, Min / Max, LRU / LFU

Implementation & Theory

Implement Stack

https://www.geeksforgeeks.org/problems/implement-stack-using-array/1

Implement Queue

https://www.geeksforgeeks.org/problems/implement-queue-using-array/1

Implement Queue using Stacks

https://leetcode.com/problems/implement-queue-using-stacks/

Implement Stack using Queues

https://leetcode.com/problems/implement-stack-using-queues/

Implement Circular Queue

https://www.geeksforgeeks.org/dsa/introduction-to-circular-queue/

Implement Deque

https://www.geeksforgeeks.org/problems/deque-implementations/1

Theory - Infix / Postfix / Prefix Expression

Questions

Valid Parentheses

https://leetcode.com/problems/valid-parentheses/

Evaluate Reverse Polish Notation

https://leetcode.com/problems/evaluate-reverse-polish-notation/

Next Greater Element I

https://leetcode.com/problems/next-greater-element-i/

Daily Temperatures

https://leetcode.com/problems/daily-temperatures/

Next Greater Element II

https://leetcode.com/problems/next-greater-element-ii/

Online Stock Span

https://leetcode.com/problems/online-stock-span/

Asteroid Collision

https://leetcode.com/problems/asteroid-collision/

Remove K Digit

https://leetcode.com/problems/remove-k-digits/description/

Largest Rectangle in Histogram

https://leetcode.com/problems/largest-rectangle-in-histogram/

Trapping Rain Water

https://leetcode.com/problems/trapping-rain-water/

Sliding Window Maximum

https://leetcode.com/problems/sliding-window-maximum/

Stack Design

Min Stack

https://leetcode.com/problems/min-stack/

Design a Stack with Increment Operation

https://leetcode.com/problems/design-a-stack-with-increment-operation/description/

LRU Cache

https://leetcode.com/problems/lru-cache/

LFU Cache

https://leetcode.com/problems/lfu-cache/

SLIDING WINDOW

Keywords:

Subarray, Substring, Contiguous, Consecutive, Window, Fixed Size, Variable Size, At Most K, At Least K, Exactly K, Longest, Shortest, Maximum, Minimum, Frequency, Frequency Map, Count, Distinct

Identifier:

Jab Subarray / Substring dikhe aur question ho:
Longest / Shortest / Maximum / Minimum / Number of
with:

- At Most K
- Exactly K
- At Least K
- Distinct Characters
- Frequency
- Count

Hint:

Jab Subarray ya Substring dikhe...
At Most K → Exactly K → At Least K → Distinct Characters → Frequency → Count

Pattern:

Right badhao → Condition check karo → Condition break ho → Left badhao

What to Look For:

Fixed-size window, Variable-size window, Longest substring, Shortest subarray, At Most K, Exactly K, Frequency, Distinct elements, Minimum valid window

Questions

Maximum Average Subarray I

`https://leetcode.com/problems/maximum-average-subarray-i/`

Maximum Number of Vowels in a Substring of Given Length

`https://leetcode.com/problems/maximum-number-of-vowels-in-a-substring-of-given-length/`

Minimum Size Subarray Sum

https://leetcode.com/problems/minimum-size-subarray-sum/

Longest Substring Without Repeating Characters

https://leetcode.com/problems/longest-substring-without-repeating-characters/

Fruit Into Baskets

https://leetcode.com/problems/fruit-into-baskets/

Longest Substring with At Most K Distinct Characters

https://leetcode.com/problems/longest-substring-with-at-most-k-distinct-characters/

Max Consecutive Ones III

https://leetcode.com/problems/max-consecutive-ones-iii/

Permutation in String

https://leetcode.com/problems/permutation-in-string/

Longest Repeating Character Replacement

https://leetcode.com/problems/longest-repeating-character-replacement/

Subarray Product Less Than K

https://leetcode.com/problems/subarray-product-less-than-k/

Number of Substrings Containing All Three Characters

https://leetcode.com/problems/number-of-substrings-containing-all-three-characters/

Subarrays with K Different Integers

https://leetcode.com/problems/subarrays-with-k-different-integers/

Minimum Window Substring

https://leetcode.com/problems/minimum-window-substring/

TWO POINTER

Keywords:

Two Pointer, Left, Right, Sorted Array, Pair Sum, Triplet, 3Sum, 4Sum, Container, Fast Pointer, Slow Pointer, In-place, Partition

What to Look For:

Sorted array, Pair sum, 3Sum, 4Sum, Target, Left / Right, Remove duplicates, Reverse, Container, Partition, Fast / Slow

Questions

Two Sum II — Input Array Is Sorted

https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/

Valid Palindrome

https://leetcode.com/problems/valid-palindrome/

Remove Duplicates from Sorted Array

https://leetcode.com/problems/remove-duplicates-from-sorted-array/

Container With Most Water

https://leetcode.com/problems/container-with-most-water/

Trapping Rain Water

https://leetcode.com/problems/trapping-rain-water/

Boats to Save People

https://leetcode.com/problems/boats-to-save-people/

Valid Triangle Number

https://leetcode.com/problems/valid-triangle-number/

Find the Duplicate Number

https://leetcode.com/problems/find-the-duplicate-number/

HEAP / PRIORITY QUEUE

Keywords:

Heap, Priority Queue, Min Heap, Max Heap, Top K, Kth Largest, Kth Smallest, Most Frequent, K Closest, Stream, Median, K-Way Merge, Scheduling

What to Look For:

Top K, Kth largest, Kth smallest, Most frequent, K closest, Continuously find min / max, Stream, Median, Merge K sorted lists, K-way merge

Implementation

Heapify

https://www.geeksforgeeks.org/problems/min-heap-implementation/1

Questions

Kth Largest Element in an Array

https://leetcode.com/problems/kth-largest-element-in-an-array/

Last Stone Weight

https://leetcode.com/problems/last-stone-weight/

Top K Frequent Elements

https://leetcode.com/problems/top-k-frequent-elements/

K Closest Points to Origin

https://leetcode.com/problems/k-closest-points-to-origin/

Kth Smallest Element in a Sorted Matrix

https://leetcode.com/problems/kth-smallest-element-in-a-sorted-matrix/

Merge K Sorted Lists

https://leetcode.com/problems/merge-k-sorted-lists/

Find K Pairs with Smallest Sums

https://leetcode.com/problems/find-k-pairs-with-smallest-sums/

Smallest Range Covering Elements from K Lists

https://leetcode.com/problems/smallest-range-covering-elements-from-k-lists/

Find Median from Data Stream

https://leetcode.com/problems/find-median-from-data-stream/

Sliding Window Median

https://leetcode.com/problems/sliding-window-median/

TOP CHANNELS TO LEARN TOPICS

Recursion - Aditya Verma , Take U Forward

Backtracking - Apna College , Pepcoding

Stack / **Queue** / **Deque** - Shradha Khapra , Take U Forward

Sliding Window & Two Pointer - Take U Forward , CTO Bhaiya

Heap - CTO Bhaiya , Coder Army

TOPIC RECOMMENDED CHANNELS & LINKS

Topic	Recommended Channels & Links
Recursion	• Aditya Verma: https://www.youtube.com/@TheAdityaVerma • Take U Forward: https://www.youtube.com/@takeuforward
Backtracking	• Apna College: https://www.youtube.com/@ApnaCollegeOfficial • Pepcoding: https://www.youtube.com/@pepcoding
Stack / Queue / Deque	• Shradha Khapra: https://www.youtube.com/@ShradhaKhapra • Take U Forward: https://www.youtube.com/@takeuforward
Sliding Window & Two Pointer	• Take U Forward: https://www.youtube.com/@takeuforward • CTO Bhaiya: https://www.youtube.com/@ctobhaiya
Heap	• CTO Bhaiya: https://www.youtube.com/@ctobhaiya • Coder Army: https://www.youtube.com/@coderarmy